

Pattern Based Loop Questions (Beginner to Very High Level)

Beginner Level

1. Square of *

```
*****
*****
*****
*****
*****
```

2. Right-angled triangle

```
*
**
***
****
*****
```

3. Inverted triangle

```
*****
****
***
**
*
```

4. Increasing number triangle

```
1
12
123
1234
12345
```

Intermediate Level

5. Pyramid of *

```
  *
 ***
*****
*****
*****
*****
```

6. Diamond

```
  *
 ***
*****
*****
*****
*****
*****
```

```
***
*
```

7. Floyd's Triangle

```
1
2 3
4 5 6
7 8 9 10
```

8. Pascal's Triangle

```
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
```

Advanced Level

9. Number Palindrome Pyramid

```
1
121
12321
1234321
123454321
```

10. Character Pyramid

```
A
ABA
ABCBA
ABCD CBA
ABCDEDCBA
```

11. Hollow Diamond

```

 *
* *
 * *
*   *
* * * *
*   *   *
* *   *   *
*   *   *
* *   *   *
*   *
 *

```

12. Butterfly Pattern

```

*           *
**         **
***       ***
****     ****
*****   *****
****     ****
**       **
*         *
```

Very High Level

13. Spiral Matrix

```
1  2  3  4
12 13 14 5
11 16 15 6
10  9  8  7
```

14. Zig-Zag Pattern

```
1 2 3
6 5 4
7 8 9
```

15. Hourglass

```
* * * * *
* * * * *
* * * *
* * *
*
* * *
* * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

16. Concentric Square

```
4444444
4333334
4322234
4321234
4322234
4333334
4444444
```

17. Heart Shape

```
* *      * *
* * * *  * * * *
* * * * * * * *
* * * * * * * *
* * * * *
* * * * *
* * * *
* * *
* *
*
```